

Final Project Proposal

Your Name Here! (group members here, optional)

Date goes Here!

Final Project Proposal

Final projects can be completed in groups of up to four students. The proposals can also be collaborative, but each student should turn in one.

Proposals should answer the following questions:

1. What is your research question? Why does it matter? How does it relate to society?
2. What data will you use to answer the question? Do you have access to these data?
 - The class list that we made together in Problem Set 2 is available below.
 - Ideally, you are able to show access to at least a portion of the data that you will need.
 - Beyond the APIs and other data sources we have used in class, there are some great options on Kaggle.
 - Consider conducting a survey! All Our Ideas is one tool that could be useful for building such a survey.
 - See an example project on wildfires from a similar class here.
3. What methods will you use to answer your research question?
 - Most projects will use some of the methods that we haven't yet covered. This is great, but show that you have an idea of where you are heading!
4. What kind of product will you create? How do you want others to read or engage with your project?
 - In most cases, projects will be R Markdown documents or Jupyter Notebooks, both of which can be published as websites.

The class list of ideas for data sources is on the course site.

```
#Libraries library(RCurl) library(readr) library(dplyr) library(tidyr) library(ggplot2)
```

```
#Our Work Here
```

```
url <- getURL("https://raw.githubusercontent.com/avaelisa/soc10finalproject-alex-christine-ava-luci/refs/heads/main/soc10-dataset")
```

download incarceration data (from github)

```
data <- read_csv(url)
```

```
#View(data)
```

```
#Mutuate to fix state names and remove U.S. total, Federal, and All States data %<>% mutate(Location=case_match(Location, "Alaska/b"~"Alaska", "Connecticut/b"~"Connecticut", "Delaware/b"~"Delaware", "Hawaii/b"~"Hawaii", "Maryland/c"~"Maryland", "Missouri/d"~"Missouri", "Oklahoma/c"~"Oklahoma", "Rhode Island/b"~"Rhode Island", "Vermont/b"~"Vermont", .default=Location)) %>% filter(!Location %in% c("U.S. total", "Federal", "All States"))
```

```

#Bar Chart 1: Total Incarceration by State #Goal: Compare incarceration rates for all states #coord_flip
will make bar chart horizontal. This will make data easier to analyze #reorder in ggplot will rank bars from
smallest to biggest so it is easier to see trends and which states have highest vs lowest incarceration rates
total_graph <- ggplot(data, aes(x = reorder(Location, Total), y = Total))+ geom_bar(position = "dodge",
stat = "identity", aes(fill = Location))+ labs(x = " ", y = "Total")+ coord_flip()

#See visualization total_graph

#Bar Chart 2: Total Incarceration Rate with States Divided by Region

#The states were divided into regions through case_match. region_data <- data %>% mu-
tate(region=case_match(Location, #Northeast c("Maine", "Vermont", "New Hampshire", "Massachusetts",
"Connecticut", "Rhode Island", "New York", "Pennsylvania", "New Jersey") ~ "Northeast", #Midwest
c("North Dakota", "South Dakota", "Nebraska", "Kansas", "Minnesota", "Iowa", "Missouri", "Illinois",
"Wisconsin", "Michigan", "Ohio") ~ "Midwest", #West c("Washington", "Oregon", "California", "Nevada",
"Idaho", "Wyoming", "Montana", "Colorado", "Utah", "Arizona", "New Mexico") ~ "West", #South
c("Texas", "Oklahoma", "Arkansas", "Louisiana", "Mississippi", "Kentucky", "Tennessee", "Alabama",
"Georgia", "West Virginia", "Florida", "South Carolina", "North Carolina", "Maryland", "Delaware") ~
"South", .default = "Other")) %>% group_by(Region) %>% mutate(sum = sum(Total, na.rm = TRUE))

#View(region_data)

region_total_graph <- ggplot(region_data, aes(x = reorder(Region, Total), y = Total)) + geom_bar(aes(fill
= Region), position = "dodge", stat = "identity")+ labs(x = "Region", y = "Total")+ theme_linedraw()+
coord_flip()

region_total_graph

#State by Gender

total_graph_gender <- ggplot(data, aes(x = reorder(Location, Total), y = Total))+ geom_bar(position =
"stack", stat = "identity", aes(fill = Location))+ labs(x = " ", y = "Total")+ coord_flip()

#test

data2 <- data %>% pivot_longer(cols = c(Male, Female), names_to = "Gender", values_to = "Value")

ggplot(data2, aes(x = Location, y = Value))+ geom_bar(position = "stack", stat = "identity", aes(fill =
Gender))+ theme_linedraw()+ labs(x = "State Separated by Gender")+ coord_flip()

#Region by Gender #Make it long

```

pivot longer

```

region_data2 <- region_data %>% pivot_longer(cols = c(Male, Female), names_to = "Gender", values_to
= "Value")

```

now try with long data

```

#View(region_data2)

```

first, we can plot a histogram of disaster years

```

gender_by_region <- ggplot(region_data2, aes(x = Region, y = Value))+ geom_bar(position = "stack",
stat = "identity", aes(fill = Gender))+ theme_linedraw()+ labs(x = "Region Separated by Gender")+
coord_flip()

gender_by_region

```

#line width? <https://r-graph-gallery.com/package/ggpattern.html> can make outside of barchart the differentiator